Abhishek Dixit, Ashish Mani, Rohit Bansal,
CoV2-Detect-Net: Design of COVID-19 prediction model based on hybrid DE-PSO with SVM using chest X-ray images,
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Abstract: For Covid-19 suspected cases, it is critical to diagnose them accurately and rapidly so that they can be isolated and provided with required medical care. A self-learning automation model will be helpful to diagnose the COVID-19 suspected individual using chest X-rays. AI based designs, which utilizes chest X-rays, have been recently proposed for the detection of COVID-19. However, these approaches are either using non-public database or having a complex design. In this study we have proposed a novel framework for real time detection of coronavirus patients without manual intervention. In our framework, we have introduced a 3-step process in which initially K-means clustering, and feature extraction is performed as a data pre-processing step. In the second step, the selected features are optimized by a novel feature optimization approach based on hybrid differential evolution algorithm and particle swarm optimization. The optimized features are then feed forwarded to SVM classifier. Empirical results show that our proposed model is able to achieve 99.34% accuracy. This shows that our model is robust and sustainable in diagnosis of COVID-19 infected individual.
Keywords: SARS-CoV-2; COVID-19; Differential evolution algorithm; Particle swarm optimization; Support Vector machine

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RESUME

Sudhanshu Jain

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Team Leader around 12 + years' experience in problem solving and implementing complex software systems. Successfully managed projects on various technical platforms and in different functional domains.

- Domain Knowledge
  - ✓ Ecommerce
- Hands on With
  - ✓ Development from scratch to live or maintain existing system using web stacks (Front end or backend based application).
  - ✓ Performance optimization for static assets using various tool like webpagetest.org
  - ✓ Build tool like Package Manager (npm), Build System Jenkins.
  - ✓ Experienced in Agile Development Environment
- Daily life
  - ✓ Coding
  - ✓ Coding review for strict quality code
  - ✓ Basic understanding of SEO
  - ✓ Good understanding of MVC architecture and client server implementation

- Worked for both Development and Maintenance software projects.
- Good knowledge of SDLC process, design, configuration control and defect management.
- Successfully managed onsite-offshore model for various projects

- Assets:
  - ✓ Open for all technologies.
  - ✓ Dynamic and effective team member
  - ✓ Good Analytical skills
  - ✓ Result Oriented.
  - ✓ Highly motivated for creative and innovative work.

Professional Experience

- Working as a Technical Architect in SDG Corporation Since , Feb 2018 to till now.
- Working as a Tech Lead in **Tech Mahindra**, since April 2013, Feb 2018
- Worked as a Senior Software Engineer in **Birlasoft Pvt. Ltd.** From May-2010 to April-2013.
- Worked as a Lead Engineer in **HCL-Technologies Ltd**. From Sep-2007 to May-2010.
- Worked as a Software Engineer in **NIIT Technologies Ltd**. From July 2006 to Aug-2007.

Technical Proficiency

Primary Skills	JavaScript, ReactJS, Redux, AngularJS 1.x, selenium, JQuery, NodeJS, Jest and Enzyme, Jasmine, Karma, Selenium, HTML5, CSS, Bootstrap, Accessibilty and material UI
Secondary Skills	Day Communique, Oracle Fusion middleware (ADF), Oracle 11 g, MySQL, core Java and J2EE (Servlet, JSP, Struts Framework) and Agile Methodology

Exposure:

Web Technologies	JavaScript, JSX, AG-Grid, ReactJS, redux JQuery, AngularJS, Twitter bootstrap, CSS, HTML, node.js, jasmine , karma, selenium and Day communiqué
Java Technologies	Java, servlet,jsp,jdbc, jndi, javabeans, struts1.2
Server	Node JS, Apache, Weblogic, Tomcat
Tools	VS Code, ATOM, Webpack, Eclipse, JDeveloper, Oracle SOA Suit, Dreamweaver, Notepad++, Microsoft office, SQL Developer, selenium, Ag-grid
Framework	ReactJS, AngularJS, Struts1.2, ADF, and IA3
Database	Mysql, oracle 9i and SQL server 10.2
Config. Mgmt. Tool	Azure, Github ,SVN, CVS
Job function	Team Management, Client Interaction, Requirement gathering, Designing and Documentation, Application and development

Professional Qualification

MASTER OF COMPUTER APPLICATION, 2005(MCA)
Faculty Of Management & Computer Application RBS College (Agra, U.P)
Affiliated to U.P Technical University, Lucknow (U.P)).

BACHELOR OF SCIENCE, 2002 (Computer Science)
D.D.I.V.E (Agra Affiliated to Dr.B.R.Ambedkar University).

PROJECT UNDERTAKING

Title VAR (Smart Sheet) and Development of Reusable components

Duration March 2019 – Till Date

Client (Confidential) **, USA**

Environment React.js, Redux, AG-Grid, JavaScript, Node.js, material design, jasmine, html5 and css.

Description

VAR

View all Records (VAR) page is used to display the submitted form data at one place as smartsheet (excel) feature, where admin and record owner(s) are able to add, edit, view and bulk update the records on the same screen along with other feature session filters, advance filters, conditional formatting, import and export data , sorting , QR code and so many other feature as like excel sheet . VAR is also help client in terms of hard saving. Now user migrate from smart sheet to VAR.

Role and contribution

- Development of Reusable components for Support Center products Involved in designing, managing and development for reusable node modules like Feedback, Global Header, footer, comments. This component used in across application in support center tech based stack portals.
- Having an ownership and responsibilities of the development of VAR product and Execution Activities and also monitoring the same.
- Framing the complete solution using AG-Grid framework and material ui design for the functionality with respect to various technology stack involved
- Interacted with clients to understand their existing applications and design the product according to their system requirements.
- Participated in day-to-day meeting, status meeting, strong reporting and effective communication with project manager and developers.
- Developed Application to asses JSON from Restful web service from consumer side using Redux and React.JS
- Developed the front-end module by using HTML5, CSS3, and JavaScript, taking advantage of the frameworks React.JS.
- Responsible for creating the JavaScript code, for different modules. JavaScript, JQuery, React.js, CSS, HTML.
- Lead Frontend team and implement project structure
- Participation in scrum meeting to discuss about project progress report

Title Ecare (Search)

Duration March 2017 – Till Date

Client **AT&T** (American Telephone and Telegraph) **, USA**

Environment React.js, Redux, JQuery, JavaScript, Node.js, JSON, fusion, Java, J2EE, Apache solr search, (NVDA) web accessibility, jasmine, karma and Selenium

Description Ecare Search project used for searching the devices, product information, accessories, support, AT&T News and frequently ask question corresponding to user searched query. The search result is displayed based on the boosting weight.

Role and contribution

- Interacted with clients to understand their existing applications and design the site according to their system requirements.
- Participated in day-to-day meeting, status meeting, strong reporting and effective communication with project manager and developers.
- Developed Application to asses JSON from Restful web service from consumer side using Redux and React.JS

- Developed the front-end module by using HTML5, CSS3, and JavaScript, taking advantage of the frameworks React.JS.
- Responsible for creating the JavaScript code, for different modules. JavaScript, JQuery, React.js, CSS, HTML.
- Lead Frontend team and implement project structure
- Having an ownership and responsibilities of the development and Execution Activities and also monitoring the same.
- Framing the complete solution for the functionality with respect to various technology stack involved
- Participation in scrum meeting to discuss about project progress report

Title Virgin Mobile FP Evolution

Duration Feb 2016 – March 2017

Client Virgin Mobile Sprint, USA

Environment React.js, JQuery, JavaScript, Axios, Node.js, J2EE, Spring Boot, Rest web services, JSON, Apache Webserver, Tomcat

Description Virgin Mobile website deals with listing out the latest phones available based on Search criteria provide by end customer. User can buy any of the phones listed on website and also buy a plan with device.

Role and contribution

- Interacted with clients to understand their existing applications and design the site according to their system requirements.
- Participated in day-to-day meeting, status meeting, strong reporting and effective communication with project manager and developers.
- Developed Application to asses JSON from Restful web service from consumer side using JavaScript and React.JS
- Developed the front-end module by using HTML5, CSS3, and JavaScript, taking advantage of the frameworks React.JS.
- Responsible for creating the JavaScript code, for different modules. JavaScript, JQuery, React.js, CSS, HTML.
- Lead Frontend team and implement project structure
- Having an ownership and responsibilities of the development and Execution Activities and also monitoring the same.
- Framing the frontend architectureS solution for the functionality with respect to various technology stack involved
- Present demo at the end of each iteration to onsite team.

Title MDV Project

Duration Jan 2015 –Jan 2016

Client **AT&T** (American Telephone and Telegraph) **, USA**

Environment Angular.js, JQuery, javascript,jsp, ATG, Weblogic, Communique5, Java, J2EE, Eclipse, notepad++,Window 2007

Description AT&T is the second largest provider of mobile telephone and the largest provider of fixed telephone in the United States, and also provides broadband subscription television services.

Market digital visualization, application is used to analytics the user driven on application. The application tracks the user activities and passes the user events/actions to the Adobe analytical tool using DTM layer. It’s an analytical tool like Webtrends.

Role and contribution

- Interacted with clients to understand their existing applications and design the site according to their system requirements.
- Participated in day-to-day meeting, status meeting, strong reporting and effective communication with project manager and developers.
- Developed Application to asses JSON from Restful web service from consumer side using JavaScript and Angular.JS
- Developed the front-end web page by using HTML5, CSS3, and JavaScript, taking advantage of the frameworks Angular.JS.
- Responsible for creating the JavaScript code, for different modules. JavaScript, JQuery, Angular.JS, CSS, HTML, XHTML
- Playing vital role for a small team and preparing daily workloads for team
- Having an ownership and responsibilities of the development and Execution Activities and also monitoring the same.
- Framing the complete solution for the functionality with respect to various technology stack involved
- Responsible for migration of components onto various servers using CVS.
- Participation in scrum meeting to discuss about project progress report
- Present demo at the end of each iteration to the clients.

Title Zodiac

Duration Aug 2013 –Dec-2014

Client AT&T (American Telephone and Telegraph) , USA

Environment Angular.js, JQuery, javascript,jsp, ATG, Weblogic, Communique5, Java, J2EE, Eclipse, notepad++,Window 2007

Description The Zodiac project is used to make the merger of other operators customer mobile plans with AT & T . Similar and upgraded plan has been provided for the merger.There is several module like authentication, device landing, service,cart summary and personal information page are used in this project.
UMP module is used for move order manual processing

Role and contribution

- Interacted with clients to understand their existing applications and design the site according to their system requirements.
- Participated in day-to-day meeting, status meeting, strong reporting and effective communication with project manager and developers.
- Developed Application to asses JSON from Restful web service from consumer side using JavaScript and Angular.JS
- Developed the front-end web page by using HTML5, CSS3, and JavaScript, taking advantage of the frameworks Angular.JS.
- Responsible for creating the JavaScript code, for different modules. JavaScript, JQuery, Angular.JS, CSS, HTML, XHTML
- Playing vital role for a small team and preparing daily workloads for team
- Having an ownership and responsibilities of the development and Execution Activities and also monitoring the same.
- Framing the complete solution for the functionality with respect to various technology stack involved
- Responsible for migration of components onto various servers using CVS.
- Participation in scrum meeting to discuss about project progress report
- Present demo at the end of each iteration to the clients.

Title TRW idWorkshop portal

Duration **Duration Aug 2011 –March-2013**

Client TRW, UK

Environment ADF(Application Development Framework),Java, J2EE, Pl/Sql,Oracle 11g,Webcenter Suite 11.1.1.4,UCM 11.1.1.4 ,SOA 11.1.1.4,OID 11g,JDEVELOPER 11.1.1.4, SQL developer, Toad, VSS, CVS,Windows XP

Description IdWorkshop Portal is an ambitious portal application for TRW which is a one stop solution for the automobile repair workshops. The portal application takes care of the Aftermarket workshops in the automobile industry. This application is sold as a product to different customers of TRW wherein each customer will have their own Branding and look and feel of the application. The portal application made use of the Webcenter Spaces 11g as the front end wherein the custom application were built in ADF and used as ADF task flows and JSR 286 portlets. It made use of the content management UCM as backend for storing content like documents and all. The application integrated with quite many third party applications either via SOA services or by External Application Connection provided by webcenter Suite 11g. The application was used by workshops to create job cards, perform actions, order parts, do the diagnosis of the vehicle, view technical information and much more. Application had a highly customized User interface corresponding to each customer. The user information was maintained in OID.

Role and contribution

- Framing the complete solution for the diagnostic functionality with respect to various technology stack involved
- Analysis of the specifications provided by the clients.
- Helping Team members in development.
- High level and low level Design Framing of the solution
- Deciding and customizing the integration strategies with the third party applications
- Designing the reusable components for the application
- Developed the most critical functionalities of the application comprising of a number of ADF screens with complex business logic.
- Responsible for migration of components onto various servers using VSS.
- Handled the change requests and enhancements

Title Vehicle Management System

Duration May-2011 to Aug -2011

Client Volkswagen, UK

Environment J2EE, PL/SQL, Struts, JSP. Servlet, iText, Quartz Scheduler, UNIX, glassfish.

Description VMS, or Vehicle Management System, is a web based application used by VWG UK and its retailers to maintain the vehicle registration details, after sales policy claims and events data. This system is configured for five brands in VWG, viz. Volkswagen, Audi, Skoda, SEAT and Volkswagen Commercial Vehicles and is accessed through Dealer Portal. The VMS application developed as a sub-component under the umbrella of the NVS program. The VMS applications caters to the functionality such as the Dealerships ability to service a vehicle as the process relies upon the dealers having access to the vehicle specification, service history and its service regime and currently this information resides within those mainframe systems

Role and contribution

- Development and Maintenance.
- SPOC and Client-Interaction
- Working at client side at UK.
- Involved in debugging and troubleshooting.

Title CAR, SAP-Tracking Tool and SAP-Security

Duration **June 2010 – April -2011**

Client SunChemical®

Environment Plum tree portal server, Weblogic 10.2, Struts 1.2, SQL Server 2000

Description

CAR:-

The capital appropriation request system (CAR) is a workflow based application which provides portal users with a mechanism to approve projects electronically, and to save information relating to the approval/rejection of a project for later reference.

SAP implementation Tracking Tool:-

SAP tracking tool project goal is to design a web portal based solution which facilitates the SAP release (implementation) across the different sites/locations of Sun Chemical Corporation. For this purpose we developed a Web Logic workshop application and plugged it into their corporate portal running on Plum tree portal environment

SAP Security:-

The SAP Security Workflow System allows a user to submit a security request to add, modify or deactivate a user in SAP. The request will be routed to an approver for the user's plant. Once approved, the request will be sent to the SMS Helpdesk system used by the SAP Application Outsourcing group to manage the execution of the request.
The following objectives have been defined for the system:

- Automate the process and reduce rework
- Reduce turnaround time to execute SAP security requests
- Improve visibility into the process
- Automatic tracking of request status

Role and contribution

- Development and Maintenance of above 3 projects.
- SPOC of above project and Client-Interaction
- Involved in debugging and troubleshooting.

Title **Merck Day Augmentation project**

Duration April 2007 – May 2010

Client **Merck & Co., Leading Pharmaceutical Company, US**

Environment Java, JSP, Servlets, JDBC, XML, JavaScript, CSS, Ajax, HTML, **Day communiqué**

Description Worked on Merck's internet & intranet sites developed in Day Communiqué. The site consists of the out of the box Components like Image Component, Rich text Editor Component, Navigation component, forms component, reference paragraph component Column control component, search component Developed Custom Components like Univadis Authentication Component, Logoff component

Role and contribution Programmer, DPA, PQA and Team handling

Title **SSPR (Self Service Plan Review)**

Duration Oct 2006 – March 2007

Client **ING, US**

Environment EJB Java, JSP, Servlet, Struts, JMS, Web Services, WSIF Framework, Business Objects, Document Archive, iText, Weblogic 8.1 and ESP.

Description The SSPR project is to assist brokers in providing services to their clients as there has been an ongoing need for functionality to create Plan Review packages. The application was divided into different layers, like presentation, proxy, web service, database and ESP.

SSPR assists Brokers in providing fund reports to their clients in PDF format. The application connects to Business Objects Server at backend and generates the requested report. The generated report is ftp to the pre defined location and then downloaded to be viewed by client in browser. The application has been developed using IA3 framework

Role and contribution Designing and Customization, Coding and Testing

Title	**Apollo2**
Duration	July 2006 – Sep 2006
Client	**ING, US**
Environment	J2EE technology, Struts, IBM Portal server 5.0, Weblogic 8.1 SP4, ANT,Axis, web services, RAD 6.0.1 and Windows2000
Description	We worked on a POC for migrating their existing application to IBM web sphere portal server and Weblogic application server. The migration was done on a customer site which provides feature like viewing details of investment made in different fund plans. The details were extracted from different databases and legacy systems using web services and enterprise beans.
Role and contribution	Designing and Customization, Coding and Testing

Title	**ConfoNet (Consumer Forum Network)**
Duration	Dec 2005 – June-2006 (During Training)
Client	**National Informatics Center (Working as Consultant for IBM)**
Environment	Core Java, J2EE (JSP, Servlet, JDBC, JNDI) Struts Framework with WebSphere Application Server 6.0 and DB2 8.2
Description	Consumer courts will use this software across India to provide cheap, speedy and simple redressal to consumer disputes. At present, there are 569 District Forums, 33 State Commissions with apex body as a National Consumer Disputes Redressal Commission (NCDRC) situated in Delhi. The software will provide a web based solution for legal data exchange, provide facilities to search knowledge (earlier court decisions), provide a single window access to Citizens / Lawyers / Judges / Department Users for information, allow secure access to judgements and provide integrated application for all legal proceedings.
Role and contribution	System design, coding and testing.

Extra Curricular Activities and Achievements:

- Achieved R&R award from Client (SDG Corporation)
- Achieved PAB award for best performance (Tech Mahindra)
- Achieved employee of the month award for excellence performance (Birlasoft)
- Achieved R&R award for best project. (HCL Technologies)
- Achieved Zero-D award for delivering SSPR project to ING client. (NIIT Technologies)

Personal Curricular

Date of Birth	JULY 20th, 1982
Gender	Male
Marital Status	Married
Nationality	Indian

Passport Details	Passport Number:- G0300096, with H1B Visa for US
Address	A 79, SF, Parsvnath Paradise, Mohan Nagar and Ghaziabad

All the mentioned above reflects only a part of me. There is certain individuality, certain core values that are also part of me but are difficult to pen down on paper.

Place: Ghaziabad

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Date:

Sudhanshu Jain